

EX No:1b	EDA DATA CLEANING

AIM

To perform exploratory data analysis and data cleaning on the **Life Expectancy dataset** by examining the dataset structure, dimensions, data types, statistical characteristics, missing values, duplicate values, country and status distributions, life expectancy patterns, correlations among numerical attributes, data preprocessing, visualization, and exporting the cleaned dataset.

ALGORITHM

1. Import the required libraries such as **Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn**.
2. Load the **Life Expectancy Data.csv** dataset into a Pandas DataFrame.
3. Remove unwanted spaces from the column names.
4. Display the first five records of the dataset.
5. Display the dataset information, shape, columns, and data types.
6. Generate descriptive statistical information for numerical attributes.
7. Identify missing values and duplicate records.
8. Remove duplicates and handle missing numerical values using median values.
9. Display country status, number of countries, and available years.
10. Analyze high, low, top, and bottom life expectancy records.
11. Calculate mean, median, mode, and standard deviation of life expectancy.
12. Analyze average life expectancy by country, year, and status.
13. Generate suitable visualizations for life expectancy and related attributes.
14. Perform correlation analysis, encoding, and normalization using **MinMaxScaler**.
15. Save the cleaned dataset and display the final dataset information.

CODE

```
import pandas as pd
```

```
# Load dataset
df = pd.read_csv("Life Expectancy Data.csv")
```

```
print("\nFirst 5 rows:")
display(df.head())
```

```
print("\nLast 5 rows:")
display(df.tail())
```

```
print("\nDataset information:")
df.info()
```

```
print("\nShape:")
print(df.shape)
```

```
print("\nColumns:")
print(df.columns.tolist())
```

```
print("\nData types:")
print(df.dtypes)
```

```
print("\nStatistical description:")
display(df.describe())

print("\nMissing values:")
display(df.isnull().sum())

print("\nStatus:")
display(df["Status"].value_counts())

print("\nUnique countries:")
print(df["Country"].nunique())

print("\nAvailable years:")
print(df["Year"].unique())

print("\nHigh life expectancy records:")
display(df[df["Life expectancy"] > 75])

print("\nLow life expectancy records:")
display(df[df["Life expectancy"] < 60])

print("\nMean life expectancy:")
print(df["Life expectancy"].mean())

print("\nMedian life expectancy:")
print(df["Life expectancy"].median())

print("\nMode of life expectancy:")
display(df["Life expectancy"].mode())

print("\nStandard deviation:")
print(df["Life expectancy"].std())

print("\nAverage life expectancy by status:")
display(df.groupby("Status")["Life expectancy"].mean())

print("\nAverage life expectancy by year:")
display(df.groupby("Year")["Life expectancy"].mean())

print("\nAverage life expectancy by country:")
country_avg = df.groupby("Country")["Life expectancy"].mean()
display(country_avg)

print("\nTop 10 countries:")
display(country_avg.nlargest(10))

print("\nBottom 10 countries:")
display(country_avg.nsmallest(10))

print("\nCorrelation:")
display(df.corr(numeric_only=True))

# CSV
df.to_csv("cleaned_life_expectancy.csv", index=False)

print("\nCSV file created successfully.")
```

OUTPUT

```

      Adult Mortality  infant deaths      Alcohol  percentage expenditure \
count      2928.000000      2938.000000  2744.000000      2938.000000
unique           NaN           NaN           NaN           NaN
top            NaN            NaN            NaN            NaN
freq           NaN            NaN            NaN            NaN
mean        164.796448        30.303948        4.602861        738.251295
std         124.292079        117.926501        4.052413       1987.914858
min           1.000000         0.000000         0.010000         0.000000
25%          74.000000         0.000000         0.877500         4.685343
50%         144.000000         3.000000         3.755000        64.912906
75%         228.000000        22.000000         7.702500       441.534144
max         723.000000       1800.000000       17.870000      19479.911610

```

```

      Hepatitis B      Measles  ...      Polio  Total expenditure \
count  2385.000000      2938.000000  ...  2919.000000      2712.000000
unique           NaN           NaN  ...           NaN           NaN
top            NaN            NaN  ...           NaN           NaN
freq           NaN            NaN  ...           NaN           NaN
mean        80.940461      2419.592240  ...      82.550188        5.93819
std         25.070016     11467.272489  ...      23.428046        2.49832
min           1.000000         0.000000  ...      3.000000         0.37000
25%          77.000000         0.000000  ...      78.000000         4.26000
50%          92.000000        17.000000  ...      93.000000        5.75500
75%          97.000000       360.250000  ...      97.000000        7.49250
max          99.000000     212183.000000  ...      99.000000       17.60000

```

	Country	Year	Status	Life expectancy	Adult Mortality	Infant deaths	Alcohol	percentage expenditure	Hepatitis B	Measles	...	Polio	Total expenditure	Diphtheria	HIV/AIDS	GDP	Population	thinness 1-19 years	thinness 5-9 years	Income composition of resources	Schooling
0	Afghanistan	2015	Developing	65.0	263.0	62	0.01	71.279624	65.0	1154	...	6.0	8.16	65.0	0.1	584.259210	33736494.0	17.2	17.3	0.479	10.1
1	Afghanistan	2014	Developing	59.9	271.0	64	0.01	73.523582	62.0	492	...	58.0	8.18	62.0	0.1	612.696514	327582.0	17.5	17.5	0.476	10.0
2	Afghanistan	2013	Developing	59.9	268.0	66	0.01	73.219243	64.0	430	...	62.0	8.13	64.0	0.1	631.744976	31731688.0	17.7	17.7	0.470	9.9
3	Afghanistan	2012	Developing	59.5	272.0	69	0.01	78.184215	67.0	2787	...	67.0	8.52	67.0	0.1	669.959000	3696958.0	17.9	18.0	0.463	9.8
4	Afghanistan	2011	Developing	59.2	275.0	71	0.01	7.097109	68.0	3013	...	68.0	7.87	68.0	0.1	63.537231	2978599.0	18.2	18.2	0.454	9.5

5 rows x 22 columns

```
Missing Values:
Country          0
Year             0
Status           0
Life expectancy  10
Adult Mortality  10
infant deaths    0
Alcohol          194
percentage expenditure  0
Hepatitis B      553
Measles          0
BMI              34
under-five deaths  0
Polio            19
Total expenditure 226
Diphtheria       19
HIV/AIDS         0
GDP              448
Population        652
thinness 1-19 years  34
thinness 5-9 years  34
Income composition of resources 167
Schooling        163
dtype: int64
```

RESULT